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Modern Data Analysis (WiSe 2026/2027)
1. Sheet

Start: Monday, 19.10
Submission: Until Sunday, 01.11 14:00 via https://tally.so/r/X5lBpL
Discussion: Monday, 02.11
Info: The worksheets should be solved using Python, in groups of 2-3 people and will be presented in the Tutorials.

Information

The worksheets and necessary toolboxes will be made available in the Lernraum "392246 Modern Data Analysis (V)". Worksheets will usually be released every two weeks on Monday, and discussed during the exercises on Monday two weeks later. In order to successfully finish the course, 60% of the available points have to be obtained. The Monday in between the release and discussion of the sheet will be used to discuss the implementation of the various algorithms presented in the lecture.

Exercise 1: Whitening

(*4 Points*)

The Whitening method is a frequently used preprocessing step and plays a major role in the SFA. Use the sheet1_1.py file. Visualize both data sets saved in dataset1.mat, apply the Whitening technique (sklearn's PCA has a whitening functionality) and illustrate the result, respectively. What happens? Which kind of information is preserved and which is lost?

Exercise 2: Slow Feature Analysis

(*6 Points*)

For the next exercise use the sheet1_2.py file. Here we load a dataset consisting of 72 images. These are snapshots of an object which is rotated slowly in front of a camera. The images have a resolution of 128×128 pixels yielding a dimensionality of $128^2 = 16384$ if represented as a vector, directly. In order to circumvent this, the images have been preprocessed with the PCA to project them down to 30 dimensions. The low-dimensional data as well as the PCA model are provided in excersise2.pkl.

(a) *(2 Pts.)* Visualize a few of the images by projecting the low-dimensional vectors back to the high-dimensional space (after the projection you can use the command reshape to transform the vector into a matrix and imshow to display the latter). Can you recognize what object it is?

(b) *(4 Pts.)* Apply the (linear) Slow Feature Analysis to the data set (use the sksfa package `https://github.com/wiskott-lab/sklearn-sfa`). Visualize the first two features as a function of the time t. What do these features characterize? What do you see if you plot these features against each other?

Exercise 3: Multiple Choice

(*0 Points*)

This exercise will not be graded, but discussed in the tutorials, multiple answers can be correct.

1. The following statements about Slow Feature Analysis (SFA) are correct:

- ☐ The following is a correct ordering of the algorithmic steps required for SFA:
  - (a) whiten data
  - (b) (nonlinear) transformation of data (e.g. using monomials)
  - (c) build time derivatives
  - (d) compute the eigenvectors of the covariance matrix of the derivatives with the smallest eigenvalues
  - (e) this induces SFA transformation
- ☐ SFA requires time series for both, training and data processing using a trained model.
- ☐ SFA is described axiomatically by four requirements. Thereby, we could drop the second one (unit variance) without any qualitative change of the results, as it only ensures a default scaling.
- ☐ After data whitening, we have features which are independent.
- ☐ Assume we use linear preprocessing only and we use SFA for the analysis of video data. Then it will identify objects which move linearly in the space.